Principles of Acoustics and **Vibration**

Single DoF Oscillator - Forced Response

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Previously on POAV...

- Focus on single DoF mass-spring system.
  - Form the building blocks for study of complex systems.
- Equation of motion,

$$F = m\ddot{x} + kx + c\dot{x} \tag{1}$$

  is a *second* order linear differential equation with constant coefficients.
- Homogenous solution (unforced, $F_e = 0$)

$$x(t) = A\cos(\omega_d t - \phi)e^{-\zeta\omega_n t} \tag{2}$$

- Amplitude A and phase ϕ depend on IC.

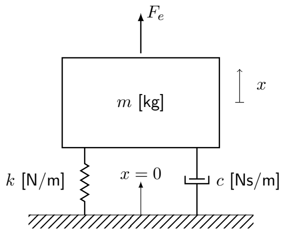


Figure: Mass-spring-damper system

Today's plan

Harmonic forcing

Forced response - no mass

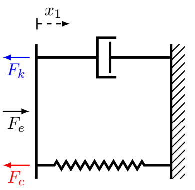


Figure: Fixed spring-dashpot system.

- First order system EoM,

$$F_e = kx + c\dot{x} \tag{3}$$

- Consider harmonic force with complex amplitude F_0 at frequency ω,

$$F_e(t) = F_0 e^{i\omega t} \tag{4}$$

- System is linear, so response will also be harmonic at ω,

$$x(t) = X_0 e^{i\omega t} \tag{5}$$

- The complex amplitude X_0 accounts for amplitude and phase changes.

Forced response - spring-dashpot (no mass)

- Using $x(t) = X_0 e^{i\omega t}$ the EoM becomes,

$$F_0 e^{i\omega t} = kX_0 e^{i\omega t} + i\omega c X_0 e^{i\omega t} \tag{6}$$

- Dividing through by $e^{i\omega t}$,

$$F_0 = kX_0 + i\omega c X_0 = \underbrace{(k + i\omega c)}_{D(\omega)} X_0 \tag{7}$$

- $D(\omega)$ is the *dynamic stiffness* of the system (N/m).
- Alternatively solving for response amplitude X_0,

$$X_0 = \frac{1}{D(\omega)} F_0 = R(\omega) F_0 \tag{8}$$

- $R(\omega)$ is called the receptance. It is a complex quantity, describes the *frequency response* of the system.

Forced response - spring-dashpot (no mass)

- Receptance can also be written in polar form with magnitude and phase,

$$R(\omega) = |R(\omega)| e^{-i\phi} \tag{9}$$

- Solution obtained by substituting in complex response amplitude,

$$x(t, \omega) = R(\omega) F_0 e^{i\omega t} \tag{10}$$
$$= |R(\omega)| F_0 e^{i(\omega t - \phi)} \tag{11}$$

- $|R(\omega)|$ determines the change in amplitude of the response relative to the input force. ϕ describes the phase shift that the system imparts.

Forced response - spring-dashpot (no mass)

- What does the magnitude $|R(\omega)|$ look like?

$$|R(\omega)| = \frac{1}{\sqrt{\text{Real}[R]^2 + \text{Imag}[R]^2}} = \frac{1}{\sqrt{k^2 + (\omega c)^2}} \tag{12}$$

- What does the phase $\phi(\omega)$ look like?

$$\phi(\omega) = \tan^{-1}\left(\frac{\text{Imag}[R]}{\text{Real}[R]}\right) = \tan^{-1}\left(\frac{c\omega}{k}\right) \tag{13}$$

Forced response - spring-dashpot (no mass)

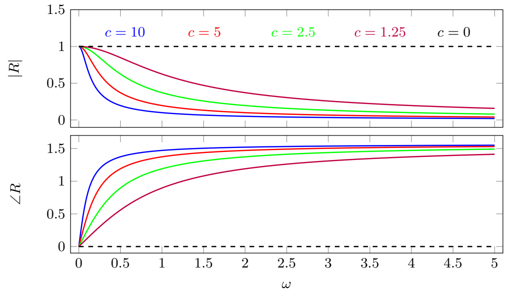


Figure: Magnitude and phase response. Fixed stiffness $k = 1$ with varying levels of damping c.

Forced response - mass-spring-dashpot

- Lets follow the same procedure including the mass term in EoM. Using $F_e(t) = F_0 e^{i\omega t}$ and $x(t) = X_0 e^{i\omega t}$,

$$F_e = kx + c\dot{x} + m\ddot{x} \qquad F_0 e^{i\omega t} = kX_0 e^{i\omega t} + ci\omega X_0 e^{i\omega t} - m\omega^2 X_0 e^{i\omega t} \tag{14}$$

- Dividing through by $e^{i\omega t}$,

$$F_0 = \overbrace{(k + i\omega c - \omega^2 m)}^{D(\omega)} X_0 \quad \rightarrow \quad X_0 = R(\omega) F_o \tag{15}$$

- What does $R(\omega)$ look like?

$$R(\omega) = \frac{1}{k + i\omega c - \omega^2 m} = \frac{1}{\underbrace{(k - \omega^2 m)}_{\text{Real}} + \underbrace{i\omega c}_{\text{Imag}}} \tag{16}$$

Forced response - mass-spring-dashpot

- Magnitude

$$|R(\omega)| = \frac{1}{|k - \omega^2 m + i\omega c|} = \frac{1}{\sqrt{(k - \omega^2 m)^2 + (\omega c)^2}} \tag{17}$$

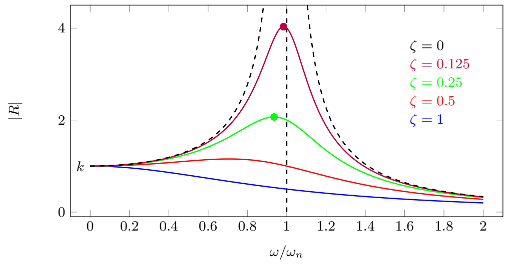
- Recall from last time $\omega_n = \sqrt{k/m}$ and $c = \zeta 2k/\omega_n$. The receptance magnitude becomes,

$$|R(\omega)| = \frac{1}{\sqrt{k^2 \left(1 - \omega^2 \frac{m}{k}\right)^2 + (\omega c)^2}} = \frac{1}{\sqrt{k^2 \left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(\zeta 2k \frac{\omega}{\omega_n}\right)^2}} \tag{18}$$

- Magnitude increases as $\left(1 - \frac{\omega^2}{\omega_n^2}\right)$ goes to zero, its maximum is limited by $\zeta 2k \frac{\omega}{\omega_n}$.

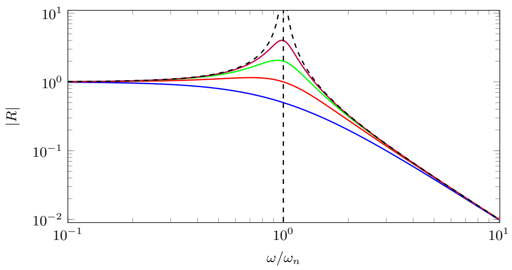
Forced response - FRF magnitude

- Peak moves to the left as damping is increased according to $\omega_{peak} = \omega_n\sqrt{1 - 2\zeta^2}$.
- If $\zeta > 1/\sqrt{2}$ response has no peak. If $\zeta = 0$, discontinuity at $\omega/\omega_n = 1$.



Forced response - FRF magnitude

- Often frequency response is plot on a log-log scale.
- Above resonance, roll-off at -12 dB/oct. Flat below resonance; stiffness controlled.



Forced response - FRF phase

- To get the phase response we need to separate real and imaginary parts,

$$R(\omega) = \frac{1}{k\left(1-\frac{\omega^2}{\omega_n^2}\right)+i\zeta 2k\frac{\omega}{\omega_n}} \times \frac{k\left(1-\frac{\omega^2}{\omega_n^2}\right)-i\zeta 2k\frac{\omega}{\omega_n}}{k(1-\frac{\omega^2}{\omega_n^2})-i\zeta 2k\frac{\omega}{\omega_n}} \tag{19}$$

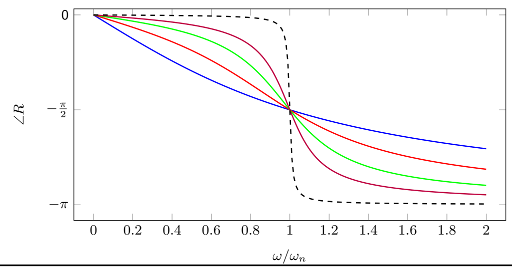
$$= \underbrace{\frac{k\left(1-\frac{\omega^2}{\omega_n^2}\right)}{k^2\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(\zeta 2k\frac{\omega}{\omega_n}\right)^2}}_{\text{real part}} - i\underbrace{\frac{\zeta 2k\frac{\omega}{\omega_n}}{k^2\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(\zeta 2k\frac{\omega}{\omega_n}\right)^2}}_{\text{imag part}} \tag{20}$$

- Phase response,

$$\angle R(\omega) = \tan^{-1}\left(\frac{\text{imag}}{\text{real}}\right) = \tan^{-1}\left(\frac{-\zeta 2k\frac{\omega}{\omega_n}}{k\left(1-\frac{\omega^2}{\omega_n^2}\right)}\right) \tag{21}$$

Forced response - FRF phase

- Motion of the mass becomes progressively delayed, relative to the excitation force.
- In the limit at $\omega \to \infty$ they are 180° out of phase.
- For all levels of damping, we have phase of $-\pi/2$ at undamped resonance ω_n.



Forced response - phase lead/lag

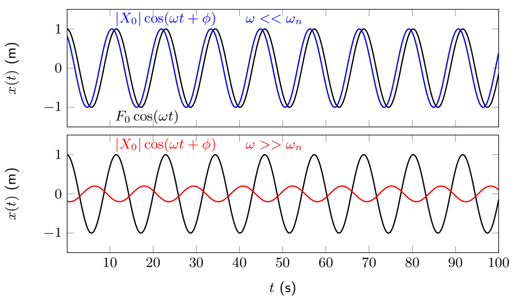


Figure: Relative phase at low (top) and high (bottom) frequencies.

Forced response at resonance (undamped)

- In the case of no damping we have discontinuity at resonance - in reality this never happens, there is *always* some damping to limit response.
- Lets look at what would happen anyway... EoM without damping,

$$m\ddot{x} + kx = F_0\cos(\omega_n t) \quad \rightarrow \quad \ddot{x} + \omega_n^2 x = \frac{F_0}{m}\cos(\omega_n t) = \frac{\omega_n^2 F_0}{k}\cos(\omega_n t) \tag{22}$$

- How do we solve this? Assume solution of $x(t) = \frac{\omega_n F_0 t}{2k}\sin(\omega_n t)$ and prove...
- First get velocity and acceleration by differentiation (using product rule),

$$\dot{x}(t) = \frac{\omega_n F_0}{2k}\sin(\omega_n t) + \frac{\omega_n^2 F_0 t}{2k}\cos(\omega_n t) \tag{23}$$

$$\ddot{x}(t) = \frac{\omega_n^2 F_0}{2k}\cos(\omega_n t) + \frac{\omega_n^2 F_0}{2k}\cos(\omega_n t) - \frac{\omega_n^3 F_0 t}{2k}\sin(\omega_n t) \tag{24}$$

Forced response at resonance (undamped)

- Plug everything into EoM and check LHS and RHS are equal...

$$\ddot{x} + \omega_n^2 x = \frac{\omega_n^2 F_0}{k}\cos(\omega_n t) \tag{25}$$

$$\overbrace{\frac{\omega_n^2 F_0}{2k}\cos(\omega_n t) + \frac{\omega_n^2 F_0}{2k}\cos(\omega_n t) - \frac{\omega_n^3 F_0 t}{2k}\sin(\omega_n t)}^{\ddot{x}} + \omega_n^2 \overbrace{\frac{\omega_n F_0 t}{2k}\sin(\omega_n t)}^{x} = \frac{\omega_n^2 F_0}{k}\cos(\omega_n t) \tag{26}$$

- It all checks out! $x(t) = \frac{\omega_n F_0 t}{2k}\sin(\omega_n t)$ is a solution at $\omega = \omega_n$. Amplitude grows linearly with time!

Forced response at resonance (undamped)

- Uncontrolled resonances have the potential to cause structural damage/failure.

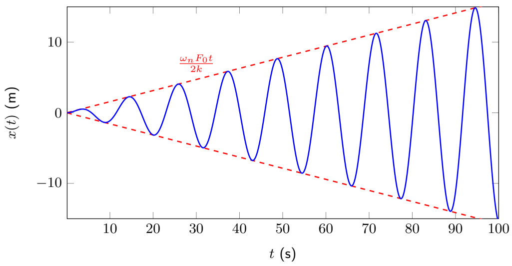
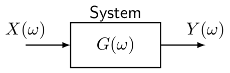


Figure: Displacement amplitude at resonance.

FRFs

Frequency response functions - FRFs



- Frequency Response Functions (FRFs) are a very general and useful concept.
- They can be obtained from models but also measurement.
- In general, an FRF describes the complex (i.e. magnitude and phase) relation between a harmonic input signal, and corresponding output signal on a given system.

$$G(\omega) = \frac{Y(\omega)}{X(\omega)} \qquad Y(\omega) = G(\omega)X(\omega) \tag{27}$$

- We can think of the FRF as a *black box* representation of the system dynamics.
- We have derived the force-displacement FRFs, dynamic stiffness $D(\omega)$ and recpetance $R(\omega)$. Velocity and acceleration-based FRFs are also common...

Displacement, velocity and acceleration-based FRFs

- For a harmonic response displacement, velocity and acceleration are related by,

$$\ddot{X}_0 = i\omega\dot{X}_0 = -\omega^2 X_0 \tag{28}$$

- Velocity-based FRFs using $\dot{X}_0 = i\omega X_0$ (impedance and mobility)

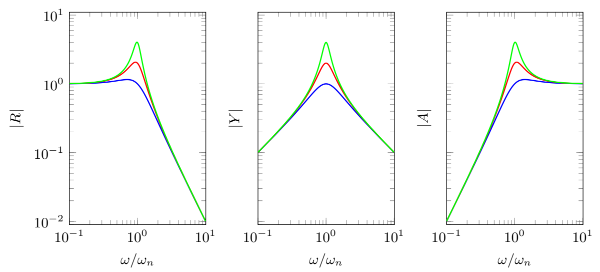
$$Z(\omega) = \frac{F_0(\omega)}{\dot{X}_0(\omega)} = \frac{F_0(\omega)}{i\omega X_0(\omega)} = \frac{1}{i\omega}D(\omega) \qquad Y(\omega) = \frac{\dot{X}_0(\omega)}{F_0(\omega)} = \frac{i\omega X_0(\omega)}{F_0(\omega)} = i\omega R(\omega) \tag{29}$$

- Acceleration-based FRFs using $\ddot{X}_0 = i\omega\dot{X}_0$ (effective mass and accelerance)

$$M(\omega) = \frac{F_0(\omega)}{\ddot{X}_0(\omega)} = \frac{F_0(\omega)}{i\omega\dot{X}_0(\omega)} = \frac{1}{i\omega}Z(\omega) \qquad A(\omega) = \frac{\ddot{X}_0(\omega)}{F_0(\omega)} = \frac{i\omega\dot{X}_0(\omega)}{F_0(\omega)} = i\omega Y(\omega) \tag{30}$$

Displacement, velocity and acceleration FRF magnitudes

- Depending on the problem, we might be interested in different response quantity, so need to use different FRF.



Arbitrary forcing

Periodic excitation

- We have covered harmonic excitation at frequency ω - periodic over time interval $T = 2\pi/\omega$.
- In vibration we often encounter signals that are periodic, but NOT necessarily harmonic.

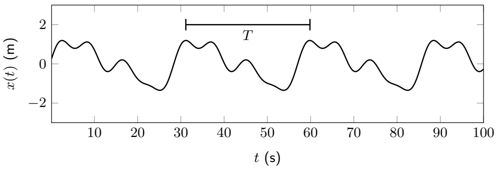


Figure: Example of periodic, but not harmonic, signal.

Periodic excitation

- Any periodic function can be represented by a convergent series of harmonic functions whose frequencies are integer multiples of a fundamental $\omega_0 = 2\pi/T$.
- This is the so-called *Fourier series*,

$$F(t) = \frac{1}{2}F_0 + \text{Real}\left[\sum_{n=1}^{\infty} F_n e^{in\omega_0 t}\right] \tag{31}$$

where F_n are complex coefficients that describe amplitude and phase of each harmonic.

- Since EoM is linear, we can treat each harmonic individually and recombine afterwards.

$$x(t) = \frac{1}{2}F_0 R(0) + \text{Real}\left[\sum_{n=1}^{\infty} R(n\omega_0) F_n e^{in\omega_0 t}\right] \tag{32}$$

Unit impulse response

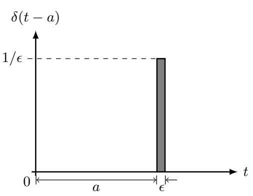


Figure: Dirac delta function.

- To consider an arbitrary, non-harmonic, forcing we can no longer assume steady state. The entire response must be regarded as transient.
- To deal with this problem, we will require the system's until impulse response.
- This can be obtained by considering the external forcing $F(t)$ to be that of a *Dirac delta* function, defined mathematically as

$$\delta(t-a) = 0 \qquad \text{for } t \neq a \tag{33}$$

$$\int_{-\infty}^{\infty} \delta(t-a)\mathrm{d}t = 1 \tag{34}$$

- Using $F_e = \delta(t)$, the system response is called impulse response $g(t)$.

Unit impulse response

- Inserting $x(t) = g(t)$ and $F_e = \delta(t)$, the EoM becomes,

$$\delta(t) = kg(t) + c\dot{g}(t) + m\ddot{g}(t) \tag{35}$$

- Taking $g(0) = \dot{g}(0) = 0$ by definition, integrate EoM over interval $\Delta t = \epsilon$ and take the limit as this goes to 0,

$$\underbrace{\lim_{\epsilon\to 0}\int_0^{\epsilon} \delta(t)dt}_{=1} = \lim_{\epsilon\to 0}\int_0^{\epsilon} \left(kg(t) + c\dot{g}(t) + m\ddot{g}(t)\right)dt \tag{36}$$

where

$$\lim_{\epsilon\to 0}\int_0^{\epsilon} m\ddot{g}(t)dt = \lim_{\epsilon\to 0} m\dot{g}(t)\Big|_0^{\epsilon} = m\lim_{\epsilon\to 0}[\dot{g}(\epsilon) - \dot{g}(0)] = m\dot{g}(0+) \tag{37}$$

- $\dot{g}(0+)$ denotes a change in velocity at the end of the increment ϵ.
- Remaining limits go to 0 - not sufficient time for displacement to develop.

Unit impulse response

- From previous equations conclude that,

$$\dot{g}(0+) = \frac{1}{m} \tag{38}$$

- Physically, unit impulse response produces an instantaneous change in velocity.
- Impulse applied at $t = 0$ is equivalent to the effect of initial velocity $\dot{x}_0 = 1/m$.
- The system impulse response is then,

$$g(t) = \frac{1}{\omega_d m}\sin(\omega_d t)e^{-\zeta\omega_n t} \qquad t > 0 \tag{39}$$

$$= 0 \qquad t < 0 \tag{40}$$

- If excitation happens at $t = \tau$, $F_e = \delta(t - \tau)$, then impulse response $g(t)$ is simply delayed by τ, $g(t - \tau)$.

Convolution integral - response to arbitrary excitation

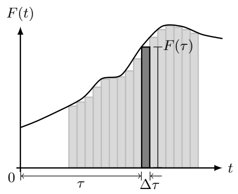


Figure: Arbitrary forcing.

- During small time instant $\Delta\tau$, we can regard $F(t)$ as an impulse $F(\tau)\Delta\tau$,

$$\Delta F(t,\tau) = F(\tau)\Delta\tau\delta(t - \tau) \tag{41}$$

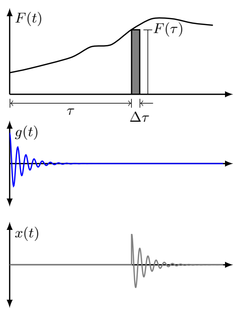
- The complete forcing function can approximated as,

$$F(t) \approx \sum F(\tau)\Delta\tau\delta(t - \tau) \tag{42}$$

- This becomes exact at $\Delta\tau \to 0$.

- Because system is linear, we can use the impulse response function $g(t)$ to predict response due to each individual excitation impulse, $\Delta F(t,\tau)$, then combine.

Convolution integral - response to arbitrary excitation



- Response due to impulse $\Delta F(t,\tau)$,

$$\Delta x(t,\tau) = F(\tau)\Delta\tau g(t - \tau) \tag{43}$$

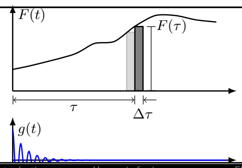
is just the impulse response $g(t)$ delayed by τ.

- The response due to full $F(t)$ is the sum of all individual impulses,

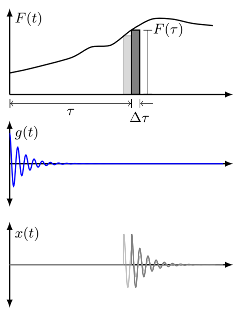
$$x(t) \approx \sum F(\tau)\Delta\tau g(t - \tau) \tag{44}$$

- This also becomes exact at $\Delta\tau \to 0$,

$$x(t) = \int_0^t F(\tau)g(t - \tau)d\tau \tag{45}$$



Convolution integral - response to arbitrary excitation



- This is called the *convolution integral* - response is expressed as a superposition of impulse responses weighted by the arbitrary forcing function.

$$x(t) = \int_0^t F(\tau)g(t - \tau)d\tau \tag{46}$$

- Known as Duhamel's integral in vibration analysis.
- Recall that convolution in the frequency domain is simply a multiplication,

$$X_0(\omega) = R(\omega)F_0(\omega) \tag{47}$$

- The FRF $R(\omega)$ is the IFT of the IRF $g(t)$.

Damping

Displacement, velocity and acceleration-based FRFs

- So far we have been assuming a viscous damping model, i.e. damping force proportional to velocity. Our EoM and FRF,

$$F_e = kx + c\dot{x} + m\ddot{x} \qquad R(\omega) = \frac{1}{k + i\omega c - \omega^2 m} \tag{48}$$

- Because of damping, system is *non-conservative* - energy is not conserved, but dissipated with time.
- The energy dissipated = work done by the external force. For a single cycle over period T (noting $\mathrm{d}x = \dot{x}\mathrm{d}t$),

$$\Delta E = \int_0^T F_e \mathrm{d}x = \int_0^{2\pi/\omega} F_e \dot{x} \mathrm{d}t \tag{49}$$

- Considering only the real parts of F_e and $\dot{x} = i\omega|R|F_0 e^{i(\omega t - \phi)}$,

$$\text{Real}(\dot{x}) = -|R|F_0\omega \sin(\omega t - \phi) \tag{50}$$

$$\text{Real}(F_e) = F_0 \cos \omega t \tag{51}$$

Energy dissipation

- The energy dissipated becomes,

$$\Delta E = \int_0^{2\pi/\omega} F_e \dot{x} \mathrm{d}t = -\int_0^{2\pi/\omega} F_0 \cos(\omega t)|R|F_0\omega \sin(\omega t - \phi)\mathrm{d}t \tag{52}$$

$$= -F_0^2\omega|R| \underbrace{\int_0^{2\pi/\omega} \cos(\omega t)\sin(\omega t - \phi)\mathrm{d}t}_{-\frac{\pi}{\omega}\sin\phi} \tag{53}$$

$$= F_0^2|R|\pi \sin\phi \tag{54}$$

- Need to determine $\sin\phi$... We know the phase angle is given by,

$$\phi = \tan^{-1}\left(\frac{\text{imag}}{\text{real}}\right) = \tan^{-1}\left(\frac{\omega c}{k - \omega^2 m}\right) \tag{55}$$

Energy dissipation

- Use the trig identity $\sin(\tan^{-1}(x)) = x/\sqrt{1+x^2}$. After a little algebra,

$$\sin\phi = \frac{\omega c}{\sqrt{(k - \omega^2 m)^2 + (\omega c)^2}} = \omega c|R| \tag{56}$$

- Energy dissipated per cycle becomes,

$$\Delta E = F_0^2|R|\pi\sin\phi = \underbrace{|R|^2F_0^2}_{|X_0|^2}\pi\omega c \tag{57}$$

- We see that the energy dissipated per cycle is proportional to: damping coefficient c, frequency ω and amplitude squared $|X_0|^2$

Displacement, velocity and acceleration-based FRFs

- In real systems, even without damping devices, energy is dissipated over time.
- Real springs (i.e. coil springs) dissipate energy as a result of internal friction within the material - as do all real structures.
- Experiments on wide variety of materials indicate that the energy loss per cycle due to internal friction is roughly proportional to amplitude squared,

$$\Delta E = \alpha |X_0|^2 \tag{58}$$

  where α is a constant *independent of frequency*.
- We call the above *structural damping* - can be attributed to hysteresis phenomenon associated with cyclic stresses in elastic materials.

Displacement, velocity and acceleration-based FRFs

- A system with structural damping can be treated as having viscous damping with an equivalent, frequency dependent, damping coefficient,

$$c_{eq} = \frac{\alpha}{\pi\omega} \quad \rightarrow \quad \Delta E = c_{eq}\pi\omega |X_0|^2 \tag{59}$$

$$= \alpha |X_0|^2 \tag{60}$$

- Our EoM becomes,

$$F_e = kx + \frac{\alpha}{\pi\omega}\dot{x} + m\ddot{x} \quad \rightarrow \quad F_0 = kX_0 + \frac{i\omega\alpha}{\pi\omega}X_0 + m\ddot{X}_0 \tag{61}$$

$$= \left(k + \frac{i\omega\alpha}{\pi\omega}\right)X_0 + m\ddot{X}_0 \tag{62}$$

- Define a 'complex stiffness' $k(1+i\gamma)$, where $\gamma = \alpha/\pi k$ is termed the *loss factor*.

Displacement, velocity and acceleration-based FRFs

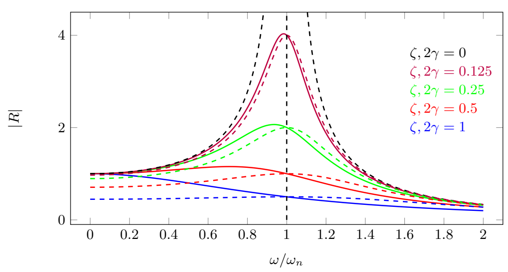
- The harmonic response then becomes,

$$x(t) = \overbrace{\frac{1}{k(1+i\gamma) - \omega^2 m}}^{R(\omega)} F_0 \cos\omega t \tag{63}$$

- **Important:** *structural damping is only valid for harmonic excitation* - we assumed $F_0 e^{i\omega t}$ in the foregoing analysis - can't use this for transient analysis!
  - Structural damping will give a *non-causal* transient response...
- Loss factors for typical materials:
  - Glass - $\sim 1 \times 10^{-4}$
  - Steel - $\sim 1 \times 10^{-3}$
  - Perspex - ~ 0.1
  - Rubber - $\sim > 0.1$

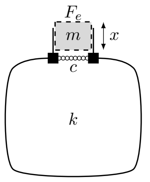
Forced response - no mass

- With structural damping, peak amplitude doesn't shift with increased damping and the response is no longer stiffness controlled at low frequency.



Some SDOF examples

Helmholtz resonator - the acoustic mass-spring-damper



- Helmholtz resonator is simply an acoustic mass-spring-damper:
  - Plug of air in neck behaves like a mass.
  - Cavity of air in volume behaves like spring.
  - Damping can be added with fine mesh.
- Mass in neck depends on cross-sectional area S, length L and air density ρ_0 (some corrections required),

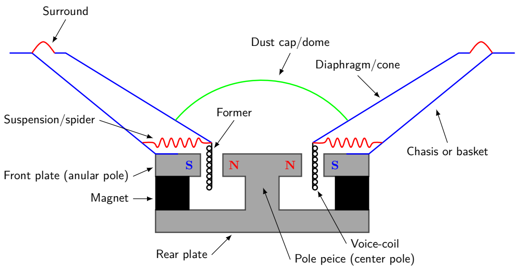
$$m = SL\rho_0 \tag{64}$$

- Stiffness of cavity depends on volume V and speed of sound c_0,

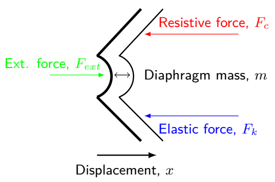
$$k = \frac{\rho_0 c_0^2 S^2}{V} \tag{65}$$

- Damping is more complicated... so just leave as c.
- External force is related to pressure over mouth $F_e = p/S$.

Loudspeaker driver - our favourite mass-spring-damper



Helmholtz resonator - the acoustic mass-spring-damper



- The suspension and spider contribute to the stiffness and damping, described by k and c.
- The mass of the diaphragm m is its mechanical mass, plus a correction for the air loading.
- The external force $F_e = BLi$ is supplied by the voice coil/magnet interaction (Lorentz force) and depends on electrical current.
- What if we put a cabinet around the back of the diaphragm?
  - We are adding an extra stiffness in parallel with that of the suspension/spider. Total stiffness becomes,

$$k_t = k_s + \frac{\rho_0 c_0^2 S^2}{V} \qquad c_t = c_s + c_v \tag{66}$$

- Radiated sound is proportional to acceleration response of surface - so interested in *accelerance FRF*.

Summary

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- If a system is being excited by a harmonic forcing, we can use the Frequency Response Function (FRF)
- FRFs can be in terms of displacement, velocity or acceleration - conversion is easy using factor of $i\omega$.
- To consider a complex periodic forcing, we can use linearity and Fourier series - simply sum up responses using FRFs.
- For arbitrary (non-period) loading we need to use convolution integral - this requires system impulse response.
- Real systems rarely possess *viscous type* damping, rather structural damping typically observed.
- We can use simple mass spring system to model Helmholtz resonators and sealed cabinet loudspeakers.

- Next time on POAV...
  - Two Degree of Freedom (DoF) system!